

I. AI Model Architecture and Performance

How does the AI model work?

ReMind.AI uses a deep convolutional neural network (CNN) based on the TinyVGG16 architecture. The pipeline accepts axial MRI slices (128x128 px, 3-channel), passes them through two convolutional blocks (each containing two Conv2d + ReLU layers followed by MaxPool2d), and classifies the result via a fully-connected head into four Alzheimer's stages: No Impairment, Very Mild, Mild, and Moderate Impairment. The model was trained on an original dataset authored by Adilan Akhramovich (Kutcher1945): approximately 11,000 real MRI slices expanded to 38,000-40,000 images using synthetic data generation (WGAN) and augmentation techniques.

What type of model is used (CNN, deep learning, etc.)?

The core model is a custom CNN (PyTorch implementation) derived from the TinyVGG16 topology: two convolutional blocks with 3x3 kernels, ReLU activations, and 2x2 max-pooling, followed by a flattened linear classifier. The architecture is deliberately lightweight to allow deployment in resource-constrained clinical environments without requiring GPU inference infrastructure.

What features does the model analyze on MRI?

The model processes raw pixel intensities of axial T1-weighted MRI slices. Convolutional filters learn spatial hierarchies automatically during training. In practice, Grad-CAM visualizations (implemented in gradcam.py) confirm that the network attends to the hippocampal region, cortical thickness reduction, ventricular enlargement, and sulcal widening -- the anatomical markers most associated with Alzheimer's progression.

How do you define 'early changes'?

Early changes correspond to the 'Very Mild Impairment' class in our dataset taxonomy, which maps to CDR 0.5 on the Clinical Dementia Rating scale. At this stage, imaging shows subtle hippocampal volume loss and mild cortical thinning in entorhinal and temporal regions, changes that are often missed on routine clinical reads but detectable by the model through population-level pattern matching.

What does '95% accuracy' mean?

The figure refers to overall classification accuracy on the held-out test set (stratified 20% split from the full dataset of 38,000-40,000 images). It means that 95% of test images were assigned the correct disease-stage label. Accuracy is reported alongside per-class precision, recall, and

F1-score in the confusion matrix outputs available in outputs/test_data_cnf_mat.png. Accuracy alone can be misleading in imbalanced settings, which is why we emphasise per-class metrics.

What are the sensitivity and specificity values?

Sensitivity (recall) and specificity are computed per class from the test-set confusion matrix. For the clinically critical Mild Impairment class, sensitivity exceeds 0.93 and specificity exceeds 0.97. For the No Impairment class (largest group) sensitivity is above 0.97. Exact per-class values are extracted from sklearn.metrics.classification_report as part of the Alzheimier_CNN.ipynb training notebook.

Is there a ROC curve?

Yes. ROC curves are generated per class using a one-vs-rest strategy with sklearn.metrics.roc_curve and roc_auc_score inside the evaluation notebook. AUC values are above 0.97 for all four classes. The curves are included in the outputs/ directory and can be provided separately on request.

How was the model validated?

Validation follows a standard supervised-learning protocol: 80/20 stratified train/test split. The model with the lowest validation loss across epochs is checkpointed (early stopping). Training and validation loss curves are saved in outputs/train-test-loss-over-epochs.png. Final evaluation is performed on the fully held-out test set, which the model never sees during training.

Are there clinical trials?

The current version has not undergone formal prospective clinical trials. Validation is performed on a curated public benchmark dataset (see Section II). We are actively designing a retrospective cohort study with a partner radiology center in Kazakhstan and are pursuing IRB approval. Clinical validation is a prerequisite before any diagnostic use in a medical context.

How do you prevent overfitting?

Several measures are applied: (1) data augmentation (random flips, rotations, brightness/contrast jitter) applied to the training split; (2) early stopping based on validation loss; (3) the TinyVGG16 architecture is intentionally small (low parameter count), reducing variance; (4) batch normalization is applied between convolutional layers in extended configurations. Train vs. validation loss convergence is monitored and visualised per epoch.

How does the model perform on real-world data?

The training dataset (authored by Adilan Akhramovich) contains approximately 11,000 real axial MRI slices expanded with WGAN-generated synthetic images to a total of 38,000-40,000. The synthetic component is generated to reproduce the statistical distribution of the real data. Performance on scanners and clinical sites not represented in the original 11,000 real slices has not yet been independently benchmarked -- this is a known limitation. Domain adaptation on Kazakhstan clinical data and a transfer-learning upgrade to ResNet-50 are planned for the next version.

Is there bias in the data (age, ethnicity, etc.)?

The real portion of the dataset (~11,000 MRI slices, authored by Adilan Akhramovich) does not include published per-patient demographic metadata, making direct quantification of age or

ethnicity bias difficult. The synthetic component (WGAN-generated) mirrors the statistical distribution of the real data and does not independently address demographic bias. The patient pool of real images is likely weighted toward Western clinical populations, which is a documented limitation for Central Asian populations. Collecting locally representative labeled MRI data from Kazakhstan is a primary objective of the planned clinical partnership.

How do you ensure explainability (interpretability for clinicians)?

We implement Gradient-weighted Class Activation Mapping (Grad-CAM) via `gradcam.py`. Grad-CAM produces a heatmap overlay on the original MRI that highlights the spatial regions most influential to the model's prediction. Clinicians can visually inspect whether the model is attending to anatomically plausible areas (hippocampus, lateral ventricles, temporal cortex) rather than image artifacts. Additionally, the Gemini-powered AI assistant in the Streamlit app translates predictions and heatmaps into natural-language clinical summaries.

II. Data Sources, Privacy, and Regulatory Compliance

How did you get access to ADNI data?

The ~11,000 real MRI slices forming the core of our dataset were sourced from ADNI (Alzheimer's Disease Neuroimaging Initiative) -- a publicly available, openly published multi-site research initiative that provides de-identified MRI data for non-commercial research use. ADNI data is accessible via adni.loni.usc.edu under a data use agreement that permits research and publication. The dataset was curated, labelled, and extended with synthetic images by Adilan Akhramovich (Kutcher1945), who is the author and rights-holder of the final composed dataset (38,000-40,000 images).

Do you have official permission to use it?

ADNI data is published under the ADNI Data Use Agreement, which permits non-commercial research use of de-identified neuroimaging data. The real MRI component (~11,000 slices) is used in full compliance with this agreement. The synthetic extension of the dataset was generated by the author (Adilan Akhramovich) using WGAN trained on the ADNI-sourced slices -- a standard research practice for dataset augmentation. No proprietary or re-identifiable patient data has been used.

Do you use local data from Kazakhstan?

Not yet in the current model version. A pilot data collection agreement with a Kazakhstan-based radiology center is under negotiation. Any locally collected MRI data will be fully de-identified in accordance with Kazakhstan Law No. 405-V 'On Health of the People and the Healthcare System' before model training or validation. We anticipate this will significantly improve model generalisability for the local population.

How do you handle personal data of patients?

The current system does not collect or process personal patient data. Users interacting with the Streamlit demo upload MRI images that are processed in memory and not persisted to disk or database. No patient identifiers (name, DOB, national ID) are requested or stored. The ReMind cognitive test platform ([remind-landing-page](#)) collects only voluntary contact information (name, phone, email) with explicit user consent, stored encrypted and used only for specialist referral purposes.

Do you comply with data protection standards (GDPR, HIPAA equivalents)?

Our data handling practices are designed to be compatible with the following frameworks:

- GDPR (EU) -- data minimisation, purpose limitation, explicit consent, right to erasure
- HIPAA (US) -- de-identification of MRI before processing, no PHI storage
- Kazakhstan Law No. 94-VI 'On Personal Data and Its Protection' (2013, amended 2021) -- data localisation requirements, consent obligations, and security standards
- ISO/IEC 27001 -- planned security framework for infrastructure upon production deployment

Formal GDPR and Kazakhstan DPA compliance audit is scheduled prior to any production launch collecting health data.

Where is data stored?

Currently all processing is stateless: uploaded MRI images are held only in application memory during inference and immediately discarded. No MRI images are written to persistent storage in the current demo. Cognitive test session data (scores, voluntary contact info) is stored in a PostgreSQL database on a dedicated server located in Kazakhstan (complying with data localisation requirements). Database access is restricted by role-based access control, encrypted at rest (AES-256), and transmitted over TLS 1.3.

Who owns the data?

The composed training dataset (ADNI-sourced real MRI slices + WGAN-generated synthetic images, total 38,000-40,000) was assembled, labelled, and published by Adilan Akhramovich (Kutcher1945). The author holds rights to the dataset as a derived work under ADNI's research use terms. The underlying raw ADNI MRI data remains the property of the ADNI initiative and its contributing institutions; redistribution of raw ADNI images is governed by the ADNI Data Use Agreement. Model weights and architecture code are proprietary assets of the ReMind project team. Any MRI data collected from Kazakhstan patients in future clinical partnerships belongs to the originating medical institution and the patient as data subject. ReMind acts as data processor, not data controller, for such patient-sourced imaging.

This document is confidential and prepared for due diligence purposes. Claims reflect the current state of the ReMind.AI prototype (April 2026). Performance figures apply to the described benchmark dataset and should not be interpreted as cleared diagnostic performance for clinical use.